

Java Programming

1. What is the name of the program that executes Java bytecode?

- A) JIT Compiler
- B) **JVM (Java Virtual Machine)**
- C) JRE (Java Runtime Environment)
- D) JDK (Java Development Kit)

Answer: B) JVM (Java Virtual Machine)

2. Which component provides the environment to run the Java program, including the JVM and libraries?

- A) JDK
- B) JIT Compiler
- C) **JRE**
- D) Interpreter

Answer: C) JRE

3. What is the extension of a compiled Java class file?

- A) `.java``
- B) `.txt``
- C) **`.class``**
- D) `.jar``

Answer: C) `.class``

4. Which command is used to compile a Java source file named `MyProgram.java``?

- A) `java MyProgram``
- B) `run MyProgram.java``
- C) `compile MyProgram.java``
- D) **`javac MyProgram.java``**

Answer: D) `javac MyProgram.java``

5. Which data type in Java is used to store single 16-bit Unicode characters?

- A) ``byte``
- B) ``short``
- C) ``int``
- D) `**`char`**`

Answer: D) ``char``

6. What is the default value of an instance variable of type ``int`` in a Java class?

- A) ``1``
- B) ``null``
- C) `**`0`**`
- D) Undefined

Answer: C) ``0``

7. Which of the following is NOT a Java keyword?

- A) ``transient``
- B) ``volatile``
- C) ``instanceof``
- D) `**`sizeof`**`

Answer: D) ``sizeof``

8. Which operator is used to determine if an object is an instance of a particular class or interface?

- A) ``is``
- B) ``typeOf``
- C) `**`instanceof`**`
- D) ``getClass``

Answer: C) ``instanceof``

9. What is the process of combining data and the methods that operate on that data into a single unit?

- A) Inheritance

- B) Polymorphism
- C) Abstraction
- D) ****Encapsulation****

Answer: D) Encapsulation

10. Which OOP principle allows an object to take on many forms?

- A) Encapsulation
- B) Abstraction
- C) Inheritance
- D) ****Polymorphism****

Answer: D) Polymorphism

11. What is the relationship between a parent class and a child class called?

- A) ****Inheritance (is-a relationship)****
- B) Aggregation (has-a relationship)
- C) Association
- D) Dependency

Answer: A) Inheritance (is-a relationship)

12. Which keyword is used to prevent a class from being inherited?

- A) ``static``
- B) ``abstract``
- C) **`**`final`**`**
- D) ``volatile``

Answer: C) ``final``

13. What is the maximum value for a variable of type ``byte``?

- A) 255
- B) 128
- C) **`**127**`**
- D) 32767

Answer: C) 127

14. What happens if a constructor is not explicitly defined in a Java class?

- A) The class will not compile.
- B) The Java Virtual Machine (JVM) calls the default constructor of the parent class.
- C) ****The compiler provides a default (no-argument) constructor.****
- D) The object is created without initialization.

Answer: C) The compiler provides a default (no-argument) constructor.

15. Which keyword is used to call a constructor of the superclass from the subclass's constructor?

- A) `this``
- B) `instanceof``
- C) `new``
- D) **`**`super`**`**

Answer: D) ``super``

16. What is the process of defining two or more methods within the same class that have the same name but different parameter lists?

- A) Method Overriding
- B) Dynamic Binding
- C) ****Method Overloading****
- D) Method Hiding

Answer: C) Method Overloading

17. Which access modifier restricts access to members only within the same class and its subclasses (regardless of package)?

- A) ``public``
- B) ``default` (package-private)`
- C) ``private``
- D) **`**`protected`**`**

Answer: D) ``protected``

18. What is the visibility of a member declared without any access modifier (known as package-private)?

- A) Visible only within the class.
- B) Visible to the class and its subclasses.
- C) `**Visible only within the package.**`
- D) Visible everywhere.

Answer: C) Visible only within the package.

19. Which statement is used to execute a block of code repeatedly as long as a condition is true?

- A) ``for``
- B) ``do-while``
- C) ``switch``
- D) `**`while`**`

Answer: D) ``while``

20. What is the correct way to declare an array of 5 integers in Java?

- A) ``int arr[5];``
- B) ``int arr[] = new int[5];``
- C) `**`int[] arr = new int[5];`**`
- D) ``array<int> arr(5);``

Answer: C) ``int[] arr = new int[5];``

21. What is the result of the expression ``5 / 2`` in integer arithmetic in Java?

- A) 2.5
- B) 3
- C) `**2**`
- D) Compiler error

Answer: C) 2

22. Which class is the root of the entire Java class hierarchy (the superclass of all other classes)?

- A) ``java.lang.Main``

- B) `java.lang.System`
- C) `java.lang.Object`
- D) `java.lang.Class`

Answer: C) `java.lang.Object`

23. What is the process of converting a primitive data type to its corresponding wrapper class object (e.g., `int` to `Integer`)?

- A) Unboxing
- B) Casting
- C) `Autoboxing`
- D) Coercion

Answer: C) `Autoboxing`

24. Which keyword is used to explicitly throw an exception from a method?

- A) `throws`
- B) `catch`
- C) `finally`
- D) `throw`

Answer: D) `throw`

25. Which block of code is guaranteed to be executed, regardless of whether an exception is thrown or caught?

- A) `try`
- B) `catch`
- C) `finally`
- D) `throw`

Answer: C) `finally`

26. What are checked exceptions?

- A) Exceptions that are always caught by the JVM.
- B) `Exceptions that must be either handled (try-catch) or declared (throws) in the method signature.`

C) Exceptions that occur only at runtime.

D) Errors that cannot be recovered from.

Answer: B) Exceptions that must be either handled (try-catch) or declared (throws) in the method signature.

27. Which keyword is used in a method signature to declare that the method might throw a certain exception?

A) `throw`

B) `catch`

C) `throws`

D) `handle`

Answer: C) `throws`

28. Which type of exception is `NullPointerException`?

A) Checked exception

B) Compile-time error

C) `Unchecked exception (Runtime Exception)`

D) Custom exception

Answer: C) `Unchecked exception (Runtime Exception)`

29. What is the key difference between an `Abstract Class` and an `Interface` in Java?

A) An Abstract Class can have static methods; an Interface cannot.

B) `An Abstract Class can have constructors and instance fields; an Interface cannot (pre-Java 8).`

C) A class can implement multiple Abstract Classes.

D) An Interface can contain concrete methods only.

Answer: B) `An Abstract Class can have constructors and instance fields; an Interface cannot (pre-Java 8).`

30. How do you declare a method in an Abstract Class that has no implementation?

A) Using the `default` keyword.

B) Using the `final` keyword.

C) **Using the `abstract` keyword.**

D) Using the `void` return type.

Answer: C) Using the `abstract` keyword.

31. What is the primary purpose of the `static` keyword when applied to a method?

A) The method can be overridden by subclasses.

B) **The method belongs to the class itself and can be called without creating an object.**

C) The method can only access static members.

D) The method is guaranteed to run in a separate thread.

Answer: B) The method belongs to the class itself and can be called without creating an object.

32. What is the difference between `==` and the `equals()` method when comparing two `String` objects?

A) `==` compares the content; `equals()` compares the memory address.

B) Both compare the content of the strings.

C) **`==` compares the memory address; `equals()` compares the content.**

D) `equals()` is used for primitive types; `==` is for objects.

Answer: C) `==` compares the memory address; `equals()` compares the content.

33. Which utility class provides static methods for sorting and searching arrays?

A) `java.util.Collection`

B) `java.lang.System`

C) `java.util.List`

D) **`java.util.Arrays`**

Answer: D) `java.util.Arrays`

34. Which interface in the Collections Framework represents a sequence of elements where duplicates are allowed and elements are ordered by index?

A) `Set`

B) `Map`

C) **`List`**

D) `Collection`

Answer: C) `List`

35. Which implementation of the `List` interface is generally better for frequent element additions and deletions from the middle of the list?

- A) `ArrayList`
- B) `Vector`
- C) *****`LinkedList`*****
- D) `Array`

Answer: C) `LinkedList`

36. Which collection class stores unique elements and maintains no specific order (implemented using a hash table)?

- A) `TreeSet`
- B) *****`HashSet`*****
- C) `LinkedHashSet`
- D) `ArrayList`

Answer: B) `HashSet`

37. Which collection interface stores key-value pairs?

- A) `Set`
- B) `List`
- C) `Collection`
- D) *****`Map`*****

Answer: D) `Map`

38. What is the process of writing an object's state to a stream, typically to a file, so it can be reconstructed later?

- A) **Serialization**
- B) Deserialization
- C) Object Mapping
- D) Persistent Storage

Answer: A) **Serialization**

39. To make an object serializable, its class must implement which interface?

- A) `java.io.SerializableObject``
- B) `java.io.ObjectStream``
- C) `**`java.io.Serializable`**`
- D) `java.io.Persistable``

Answer: C) `java.io.Serializable``

40. What is the output of the following Java code?

<!-- end list -->

```
```java
```

```
int x = 10;
```

```
System.out.println(x++);
```

```
```
```

- A) 11
- B) `**10**`
- C) 9
- D) Compiler error

Answer: B) 10

41. What is the purpose of the `**Garbage Collector**` in Java?

- A) To manage memory allocated using the ``new`` operator.
- B) `**To automatically free memory of objects that are no longer referenced by the program.**`
- C) To destroy objects using the ``delete`` keyword.
- D) To handle exceptions that occur at runtime.

Answer: B) To automatically free memory of objects that are no longer referenced by the program.

42. How can you define a constant variable in Java?

- A) ``static const int MAX_VALUE = 100;``
- B) ``final const int MAX_VALUE = 100;``
- C) `***static final int MAX_VALUE = 100;***`
- D) ``const int MAX_VALUE = 100;``

Answer: C) ``static final int MAX_VALUE = 100;``

43. Which access modifier is the most restrictive?

- A) ``protected``
- B) ``default` (package-private)`
- C) `***private***`
- D) ``public``

Answer: C) ``private``

44. Which method is the entry point for a standalone Java application?

- A) ``start()``
- B) ``run()``
- C) `***public static void main(String[] args)***`
- D) ``main(String[] args)``

Answer: C) ``public static void main(String[] args)``

45. The ``String`` class in Java is:

- A) ``mutable`` (can be changed after creation).
- B) `***immutable`` (cannot be changed after creation).`***`
- C) A primitive data type.
- D) An abstract class.

Answer: B) ``immutable`` (cannot be changed after creation).

46. Which class should be used when a String needs to be modified frequently in a single thread?

- A) ``String``
- B) `***StringBuilder***`
- C) ``StringTokenizer``

D) ``StringPool``

Answer: B) ``StringBuilder``

47. What does the ``this`` keyword refer to in a non-static method?

A) The parent class object.

B) The class definition itself.

C) `**The current object for which the method is called.**`

D) The object passed as the first argument.

Answer: C) The current object for which the method is called.

48. Which exception is thrown when attempting to access an array element with an index that is out of bounds?

A) ``IOException``

B) ``ArrayOutOfBoundsException``

C) `**`ArrayIndexOutOfBoundsException`**`

D) ``IndexOutOfBoundsException``

Answer: C) ``ArrayIndexOutOfBoundsException``

49. What is the primary purpose of Java `**Generics**`?

A) To enforce runtime type checking.

B) `**To enable type-safe code for collections and classes, with type checking done at compile time.**`

C) To allow dynamic method dispatch.

D) To create dynamic arrays.

Answer: B) To enable type-safe code for collections and classes, with type checking done at compile time.

50. What is a `**Lambda Expression**` (Java 8)?

A) A complex template specialization.

B) A new type of loop construct.

C) `**A short block of code that takes parameters and returns a value (used for implementing functional interfaces).**`

D) A type of anonymous inner class that can be stored in a variable.

Answer: C) A short block of code that takes parameters and returns a value (used for implementing functional interfaces).

51. What is the functional interface used to represent a method that accepts a single input argument and returns no result?

A) `Function<T, R>`

B) `Supplier<T>`

C) `Predicate<T>`

D) `Consumer<T>`

Answer: D) `Consumer<T>`

52. What is the function of the `java.util.stream.Stream` API (Java 8)?

A) To manage file I/O operations.

B) To manage memory streams.

C) `To process sequences of elements in a functional manner (e.g., filter, map, reduce).`

D) To handle network connections.

Answer: C) To process sequences of elements in a functional manner (e.g., filter, map, reduce).

53. Which of the following is an example of `compile-time polymorphism` (static polymorphism)?

A) Method Overriding

B) Dynamic Dispatch

C) `Method Overloading`

D) Interface Implementation

Answer: C) Method Overloading

54. The `hashCode()` and `equals()` methods are commonly overridden together. Where are they originally defined?

A) `java.lang.String`

B) `java.lang.System`

C) `java.lang.Object`

D) `java.lang.Cloneable`

Answer: C) `java.lang.Object`

55. What does the `==` operator return when comparing two arrays in Java?

- A) `true` if all elements are equal.
- B) `true` if the lengths are equal.
- C) `true` if both references point to the same array object in memory.
- D) `true` if the hash codes are equal.

Answer: C) `true` if both references point to the same array object in memory.

56. Which statement is used to force an immediate exit from the innermost loop (for, while, or do-while)?

- A) `continue;`
- B) `return;`
- C) `break;`
- D) `exit;`

Answer: C) `break;`

57. When a method overrides a superclass method, the subclass method's access modifier must be:

- A) More restrictive than the superclass method.
- B) Less restrictive or the same as the superclass method.
- C) Always `public`.
- D) Always `private`.

Answer: B) Less restrictive or the same as the superclass method.

58. Which thread state indicates that a thread is waiting to acquire a lock to enter a synchronized block/method?

- A) `RUNNABLE`
- B) `WAITING`
- C) `BLOCKED`
- D) `TERMINATED`

Answer: C) `BLOCKED`

59. What is the purpose of the `synchronized` keyword in Java multi-threading?

- A) To ensure that a method or block is always executed by the main thread.
- B) To make a variable thread-safe.
- C) ****To ensure that only one thread can execute a method or block at a time (mutual exclusion).****
- D) To force a thread to sleep for a certain duration.

Answer: C) To ensure that only one thread can execute a method or block at a time (mutual exclusion).

60. Which interface must a class implement to create a custom thread?

- A) `java.lang.Threadable`
- B) `java.util.Concurrency`
- C) `java.lang.Callable`
- D) ****`java.lang.Runnable`****

Answer: D) `java.lang.Runnable`

61. What is the output of the following Java code?

<!-- end list -->

```
```java
int a = 7;
int b = 3;
System.out.println(a % b);
```
```

- A) 0
- B) 1
- C) 2
- D) 3

Answer: B) 1

62. What is the name of the file that contains Java class libraries, resources, and metadata, usually compressed?

- A) Class file
- B) Source file
- C) **JAR file (Java Archive)**
- D) Executable file

Answer: C) JAR file (Java Archive)

63. Which package contains the fundamental classes and interfaces that are necessary for all Java programs?

- A) `java.util`
- B) **`java.lang`**
- C) `java.io`
- D) `java.main`

Answer: B) `java.lang`

64. If a method does not specify a return type, what is assumed?

- A) It returns `int`.
- B) It returns `boolean`.
- C) **The method signature is invalid (must specify `void` or a type).**
- D) It returns `Object`.

Answer: C) The method signature is invalid (must specify `void` or a type).

65. What is the default return value of a method declared as `public int calculate(int x)` if it reaches the end without a `return` statement?

- A) 0
- B) `null`
- C) The value of `x`.
- D) **Compile-time error.**

Answer: D) Compile-time error.

66. What is the use of the `break` statement in a `switch` statement?

- A) To stop the program execution.
- B) ****To exit the `switch` block after a case is matched.****
- C) To skip the current iteration of the loop.
- D) To call the default case.

Answer: B) To exit the `switch` block after a case is matched.

67. Which of the following data types can hold the largest range of integer values?

- A) `short`
- B) `int`
- C) `long`
- D) ****`double`****

Answer: C) `long` (although `double` holds the largest range overall, `long` is the largest integer type).

68. What is the output of the following Java code?

<!-- end list -->

```
``java
for (int i = 0; i < 5; i++) {
    if (i == 3) continue;
    System.out.print(i);
}
``
```

- A) 01234
- B) ****0124****
- C) 012
- D) 4

Answer: B) 0124

69. What is a **Daemon Thread** in Java?

- A) A thread with higher priority.
- B) A thread that is guaranteed to finish execution.
- C) **A thread that runs in the background and does not prevent the JVM from exiting.**
- D) A thread used only for network connections.

Answer: C) A thread that runs in the background and does not prevent the JVM from exiting.

70. Which class is used to read data from a file character by character?

- A) `FileInputStream`
- B) `ObjectInputStream`
- C) **`FileReader`**
- D) `BufferedReader`

Answer: C) `FileReader`

71. Which I/O class is used to read text from a character-input stream, buffering characters to provide for the efficient reading of characters, arrays, and lines?

- A) `Scanner`
- B) `FileReader`
- C) `InputStreamReader`
- D) **`BufferedReader`**

Answer: D) `BufferedReader`

72. What is the primary function of the **JDBC** (Java Database Connectivity) API?

- A) To establish network connections.
- B) To handle XML parsing.
- C) **To define how a Java application can interact with a database.**
- D) To manage object serialization.

Answer: C) To define how a Java application can interact with a database.

73. What is the process of loading a class into memory by the JVM called?

- A) Instance creation

- B) ****Class loading****
- C) Class referencing
- D) Bytecode execution

Answer: B) Class loading

74. What is the difference between a ****Local Variable**** and an ****Instance Variable****?

- A) Local variables can be static; instance variables cannot.
- B) Instance variables must be initialized; local variables are given default values.
- C) ****Local variables are declared inside a method and stored on the stack; Instance variables are in the class and stored on the heap.****
- D) Local variables can be accessed by any method; instance variables cannot.

Answer: C) Local variables are declared inside a method and stored on the stack; Instance variables are in the class and stored on the heap.

75. Which feature of Java prevents memory corruption by prohibiting direct pointer manipulation?

- A) Garbage Collection
- B) Platform Independence
- C) ****Absence of pointers (in the C/C++ sense)****
- D) Multithreading

Answer: C) Absence of pointers (in the C/C++ sense)

76. What is the correct way to handle a scenario where multiple exceptions might occur?

- A) Use a single ``catch`` block that catches ``Exception``.
- B) Use multiple ``catch`` blocks in any arbitrary order.
- C) ****Use multiple ``catch`` blocks, ordered from the most specific exception to the most general.****
- D) Use only ``throws`` in the method signature.

Answer: C) Use multiple ``catch`` blocks, ordered from the most specific exception to the most general.

77. What is an ****Inner Class****?

- A) A class whose methods are all ``private``.
- B) A class that has no constructors.

- C) **A class defined within another class.**
- D) A class that is only visible within its own package.

Answer: C) A class defined within another class.

78. A **Static Inner Class** (Nested Class) can access which members of the outer class?

- A) All members (static and instance).
- B) Only `public` members of the outer class.
- C) **Only `static` members of the outer class.**
- D) No members of the outer class.

Answer: C) Only `static` members of the outer class.

79. What is the difference between `StringBuffer` and `StringBuilder`?

- A) `StringBuffer` is faster; `StringBuilder` is slower.
- B) **`StringBuffer` is synchronized (thread-safe); `StringBuilder` is not (faster).**
- C) `StringBuilder` is immutable; `StringBuffer` is mutable.
- D) `StringBuffer` is the newer class (Java 8+).

Answer: B) `StringBuffer` is synchronized (thread-safe); `StringBuilder` is not (faster).

80. Which of the following is true about **Marker Interfaces** (like `Serializable` or `Cloneable`)?

- A) They contain abstract methods that must be implemented.
- B) They are used to create nested classes.
- C) **They contain no methods and are used to provide metadata or capability to the JVM.**
- D) They can only be implemented by abstract classes.

Answer: C) They contain no methods and are used to provide metadata or capability to the JVM.

81. What is the purpose of the `transient` keyword?

- A) To make a variable thread-safe.
- B) To make a variable read-only.
- C) **To prevent a member field from being serialized when the object is written to a stream.**
- D) To define a variable that changes value frequently.

Answer: C) To prevent a member field from being serialized when the object is written to a stream.

82. Which method of the `Object` class is used to create a field-by-field copy of an object?

- A) `copy()`
- B) `replicate()`
- C) `clone()`
- D) `new()`

Answer: C) `clone()`

83. What happens if you try to `clone()` an object whose class does not implement the `Cloneable` interface?

- A) The JVM performs a deep copy.
- B) The clone operation succeeds silently.
- C) A `CloneNotSupportedException` is thrown.
- D) A compile-time error occurs.

Answer: C) A `CloneNotSupportedException` is thrown.

84. What is the output of the following Java code?

<!-- end list -->

```
```java
String s1 = "hello";
String s2 = "hello";
System.out.println(s1 == s2);
```
```

- A) `false`
- B) `true`
- C) Compiler error
- D) Runtime error

Answer: B) `true` (Due to String constant pool optimization)

85. Which component of the JDK is responsible for optimizing the Java bytecode into native machine code during runtime?

- A) JVM
- B) Interpreter
- C) **JIT Compiler**
- D) Class Loader

Answer: C) JIT Compiler

86. What is the term for a class that contains only private constructors and static methods, and cannot be instantiated?

- A) Abstract class
- B) Singleton class
- C) **Utility class**
- D) Inner class

Answer: C) Utility class

87. Which method of the `Thread` class is used to manually pause the execution of the current thread for a specified duration?

- A) `yield()`
- B) `wait()`
- C) `suspend()` (deprecated)
- D) **`sleep()`**

Answer: D) `sleep()`

88. The `wait()`, `notify()`, and `notifyAll()` methods can only be called from:

- A) The main thread.
- B) Static methods.
- C) **A synchronized block or method (on the object whose lock is held).**
- D) Any public method.

Answer: C) A synchronized block or method (on the object whose lock is held).

89. Which design pattern ensures that a class has only one instance and provides a global point of access to it?

- A) Factory
- B) Observer
- C) Decorator
- D) ****Singleton****

Answer: D) Singleton

90. Which of the following data structures is guaranteed to maintain the order of insertion?

- A) `HashSet`
- B) `TreeMap`
- C) `ArrayList`
- D) **`**LinkedHashSet**`**

Answer: D) `LinkedHashSet`

91. Which wrapper class is used for the primitive type `boolean`?

- A) `Bool`
- B) `BooleanType`
- C) **`**Boolean**`**
- D) `boolean`

Answer: C) `Boolean`

92. What is the process of converting a wrapper class object back to its primitive data type (e.g., `Integer` to `int`)?

- A) Autoboxing
- B) Casting
- C) Coercion
- D) **`**Unboxing**`**

Answer: D) Unboxing

93. What is the result of the logical expression `(true && false) || (false || true)`?

- A) `false`

B) `true`

C) Compiler error

D) Runtime error

Answer: B) `true`

94. What is the name of the memory area used for storing static variables and class-level information?

A) Heap

B) Stack

C) `Method Area (or Class Area)`

D) Program Counter Register

Answer: C) `Method Area (or Class Area)`

95. In Java, all exceptions are ultimately derived from which class?

A) `java.lang.RuntimeException`

B) `java.lang.Error`

C) `java.lang.Throwable`

D) `java.lang.Catchable`

Answer: C) `java.lang.Throwable`

96. Which method is used to start the execution of a thread?

A) `run()`

B) `execute()`

C) `init()`

D) `start()`

Answer: D) `start()`

97. When working with streams, which of the following is considered a `terminal operation` (causes the stream to be evaluated)?

A) `filter()`

B) `map()`

C) `peek()`

D) `**`collect()`**`

Answer: D) ``collect()``

98. Which keyword is used to import a class from a different package?

A) ``include``

B) ``package``

C) `**`import`**`

D) ``uses``

Answer: C) ``import``

99. What is the concept of hiding the implementation details and only showing the essential features of an object?

A) Encapsulation

B) Inheritance

C) Polymorphism

D) `**Abstraction**`

Answer: D) Abstraction

100. What is the maximum number of arguments a Java method can take?

A) 64

B) 128

C) 255

D) `**No strict limit, constrained by memory/JVM limits, but practically unlimited by the specification.**`

Answer: D) No strict limit, constrained by memory/JVM limits, but practically unlimited by the specification.